LineGuard

LineGuard

2026-2027

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LineGuard

OPGW

Java Spring Boot

SSE

Java

Spring Boot

1 引言

1.1 选题背景

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1.2 机器人构想与运行场景

LineGuard Robot Dock OPGW Dock V

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1.3 本文工作

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2 机器人软件功能设计

2.1 系统组成与运行环境

Dock

2.2 软件功能需求与承载模块

2-1

2-1

功能类别	具体能力	承载模块

2.3 核心任务闭环与异常处置

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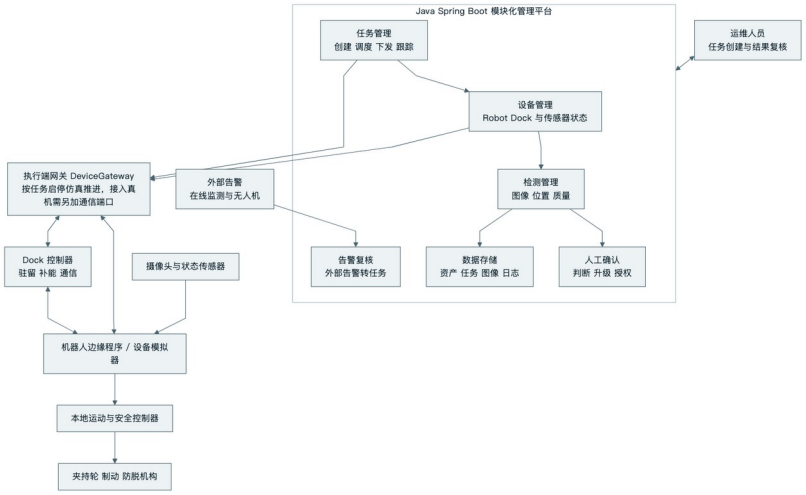

3 软件架构设计

3.1 总体分层与模块划分

Java Spring Boot

3-1

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3.2 任务状态机

3-2

3-2

Dock

3.3 数据模型

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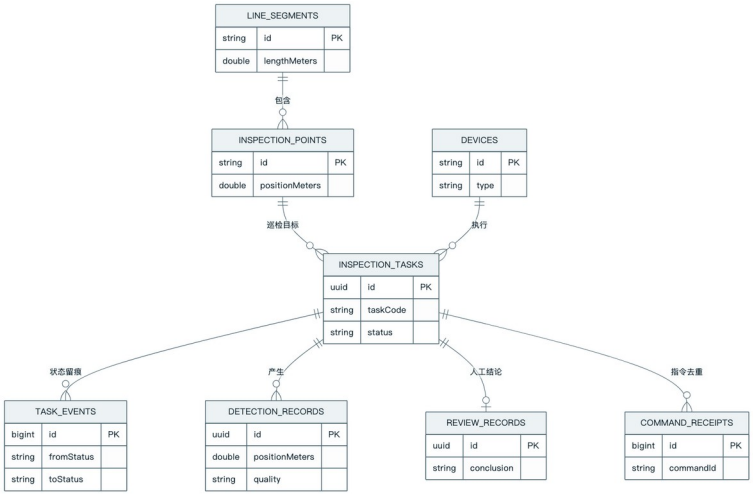

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3.4 □□□□□□□□□

" " SSE

SSE EventSource

4 关键代码实现

4.1 任务状态机

com.lineguard.service.TaskStateMachine

```
public TaskStateMachine() {
    allow(TaskStatus.CREATED, TaskStatus.PRECHECKING, TaskStatus.ABORTED);
    allow(TaskStatus.PRECHECKING, TaskStatus.DEPARTING, TaskStatus.ABORTED);
    //
}

/**                          */
public void check(TaskStatus from, TaskStatus to) {
    if (!transitions.getOrDefault(from, Set.of()).contains(to)) {
        throw new BusinessException(HttpStatus.CONFLICT,
                "              " + from + "        " + to);
    }
}
```

4.2 执行端网关与仿真推进

"   "

com.lineguard.simulation.SimulationEngine

```
private void tick(UUID taskId) {
    try {
        if (missionStateService.advanceSimulation(taskId).pausesSimulation()) {
            stop(taskId);
        }
    } catch (Exception exception) {
        //
        missionStateService.handleSimulationFailure(taskId, exception.getMessage());
        stop(taskId);
    }
}
```

4.3 □□□□

com.lineguard.service.MissionStateService

```
private boolean acceptCommand(InspectionTask task, String commandId, CommandType type) {
    if (commandRepository.existsByTaskIdAndCommandId(task.getId(), commandId)) {
        recordEvent(task, "DUPLICATE_COMMAND", "                " + commandId);
        return false;
    }
    commandRepository.save(new CommandReceipt(task.getId(), commandId, type));
    return true;
}
```

4.4 □□□□□□□□□□

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com.lineguard.service.StartupRecoveryService

```
taskRepository.findAll().stream()
        .filter(task -> TaskStatus.activeStatuses().contains(task.getStatus()))
        //
        .filter(task -> task.getStatus() != TaskStatus.WAITING_MANUAL
                && task.getStatus() != TaskStatus.SAFE_PARKED)
        .forEach(task -> task.transitionTo(TaskStatus.WAITING_MANUAL,
                "                                        "));
```

5 运行验证与结论

5.1 演示闭环

Dock

5.2 测试覆盖

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5.3 局限与结论

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